

Predicate Decomposition in Mandarin Resultative Verb Compounds Through Analysis of 打-X (Hit-X) and 踢-X (Kick-X)

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Introduction

Mandarin Chinese is a Sino-Tibetan language spoken widely across China and the Chinese diaspora. Verb-Verb compounds is a common phenomenon in Mandarin that have different functions depending on how the two verbs are related and work with each other. Resultative verb compound (RVC) is an example of such verb-verb compounds where a V2 (the second verb) describes the result of the action described by V1 (the first verb). They can be further categorized into subject-oriented resultatives (example 1) and object-oriented resultatives (example 2).

1. 他 喝醉了
tā hē-zuì-le
he drink-drunken-PERF
'He got drunk.'
2. 山姆 打破了 玻璃
shānmǔ dǎ-pò-le bō-li
Sam hit-break-PERF glass
'Sam broke the glass.'

In simple Mandarin syntax, RVCs are usually considered as a single verb compound in morphosyntactic analysis. Other scholars also proposed alternative analysis that aim to fully decomposed RVCs that will be useful to analyzing the semantic implication of the RVCs. The unique and complex features of RVCs ask for a more complex semantic explanation on the predicates they generated. This paper examines the semantics of Mandarin object-oriented RVCs through analyzing those formed with 打-X [dǎ-X] (Hit-X) and 踢-X [tī-X] (Kick-X).

Literature review

To understand the semantics of RVCs, it is also important to understand the different morphosyntactic analysis of them. Based on Thompson's analysis (1973) of RVCs, the simplest analysis could be seen in Fig 1, where 打破 dǎ-pò (hit-break) is just seen as one verb compound.

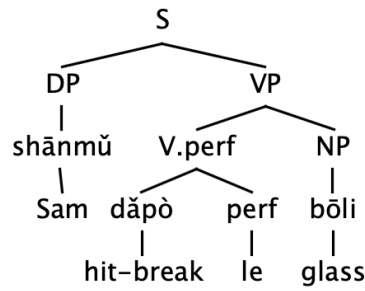


Fig 1. Simple Tree for Sentence 2

Hashimoto (1966) proposed an alternative transformation that could apply to some of the RVCs (including the examples I’m using) that separates the two verbs to form two sentences that underlies the original structure as show in Fig 2. where the raising of V2 causes pruning of the embedded VP and S nodes, which also causes the NP to be deleted as there already exist an identical NP.

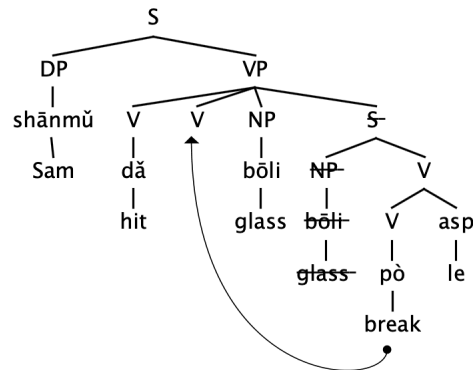


Fig 2. Alternative Tree for Sentence 2

Li (2013) offers a lexical-semantic account of Mandarin RVCs and argues that the complex thematic relations in these compounds arise from interactions between the meanings of the individual components and the meaning of the compound as a whole. This is important for my project because it supports the idea that a compound such as dǎpò can function as a single predicate in syntax while still containing multiple semantic components.

A broader literature on predicate decomposition provides the theoretical basis for analyzing causative and resultative meanings through smaller semantic predicates such as CAUSE BECOME. Dowty (1979) is an important foundation for this approach because he connects

lexical decomposition with formal semantics and discusses recurring atomic predicates such as CAUSE BECOME, which are used to represent causative verb meanings. Rappaport Hovav and Levin (1998) further develop this idea by arguing that verb meanings can be represented through articulated lexical-semantic structures rather than as unanalyzed atomic predicates; this is especially useful for verbs or constructions that combine an action with a resulting state. Wechsler and Noh (2001) extend the discussion to resultative predicates, arguing that English and Korean resultatives follow from general principles of predication and argument saturation rather than construction-specific rules. Tham (2009) applies this kind of resultative analysis directly to Mandarin, arguing that the result predicate in Mandarin RVCs encodes the change-of-state semantics of the compound; for example, in compounds like 擦干 cā-gān “wipe-dry,” the result component contributes the achieved dry state. Together, these works support a decompositional approach in which causative-resultative meanings can be represented as involving an action predicate linked to a result-state predicate through a relation such as CAUSE BECOME.

Research questions

The paper mainly explores the following research questions:

1. Are Mandarin resultative compounds such as 打破 [dǎ-pò] (Hit-break), 踢破 [tī-pò] (Kick-break), 打翻 [dǎ-fān] (Hit-overturn), and 踢翻 [tī-fān] (Kick-overturn) best represented as atomic predicates, or as complex predicates composed of an action predicate and a result predicate?
2. What semantic contribution is made by the first verb dǎ or tī, and what contribution is made by the result complement, such as 破 [pò] (break), 碎[suì] (shatter), 翻[fān] (overturn), 倒[dǎo] (fall over), or 飞[feī] (fly away)?
3. How do entailment patterns show that the result state is contributed by the complement rather than by the action verb alone?

4. How can Dowty-style predicate decomposition account for the meaning of Mandarin RVCs? And how does would the internal meaning of Mandarin RVCs be represented compositionally using lambda calculus?

Method for data collection

My data collection method combines native-speaker judgment, constructed minimal pairs, and comparison with examples discussed in the literature. Since this paper investigates Mandarin, my primary evidence consists of Mandarin sentence-level data that I'm able to gather and record as a native speaker of Mandarin. So far, I have collected a set of Mandarin resultative compounds involving *dǎ* (hit) and *tī* (kick) followed by result complements. These examples are chosen because the first verb changes while the result complement remains constant. This makes it possible to test whether the result complement has a stable semantic contribution across different compounds. The main diagnostic I use is entailment contradiction. If a compound entails a result state, then denying that result state should produce a contradiction or at least a very infelicitous sentence.

Data presentation and description

The dataset consists of 8 constructed Mandarin examples. These examples are designed to test whether Mandarin resultative verb compounds are best analyzed as atomic predicates or as semantically complex predicates built from an action predicate, a causal relation, and a result-state predicate.

First, simple action verbs do not entail result states:

- (1) 我 打了 杯子, 但是 杯子 没 破
wǒ dǎ-le bēizi, dànshì bēizi méi pò
I hit.PERF cup, but cup NEG break

I hit the cup, but the cup did not break.

- (2) 我 踢了 杯子, 但是 杯子 没 碎
wǒ tī-le bēizi, dànshì bēizi méi suì

I kick.PERF cup, but cup NEG shatter
 I kicked the cup, but the cup did not shatter.

Second, resultative compounds entail the result state contributed by the second verbal element:

(3)* 我 打破了 杯子 但是 杯子 没 破
 wǒ dǎ-pò-le bēizi, dànshì bēizi méi pò
 I hit-break.PERF cup, but cup NEG break
 I hit-break the cup, but the cup did not break.

(4)* 我 踢碎了 杯子 但是 杯子 没 碎
 wǒ tī-suì-le bēizi, dànshì bēizi méi pò
 I kick-shatter.PERF cup, but cup NEG break
 I kicked-shatter the cup, but the cup did not shatter.

These examples are contradictory or highly infelicitous. They show that the result complement contributes an entailed result state. The contrast between examples 1–2 and 3–4 supports a predicate-decomposition analysis: the meaning of the compound is not contributed by the first verb alone. Instead, the compound contains both an action component and a result component.

Third, resultative compounds encode a causal relation between the action and the result:

(5) 我 打了 杯子, 杯子 后来 破了 但是 不是 因为 我 打 它 才 破的
 wǒ dǎ-le bēizi, bēizi hòulái pò-le dànshì bùshì yīnwèi wǒ dǎ tā cái pò-de
 I hit. cup, cup later break. but NEG because I hit it then broken
 PERF PERF
 I hit the cup, and the cup later broke, but it did not break because I hit it.

(6) 我 踢了 杯子, 杯子 后来 掉了 但是 不是 因为 我 踢 它 才 掉的
 wǒ tī-le bēizi, bēizi hòulái diào-le dànshì bùshì yīnwèi wǒ tī tā cái diào-de
 I kick. cup, cup later fall. but NEG because I kick it then fallen
 PERF PERF
 I kick the cup, and the cup later fell, but it did not fall because I kicked it.

(7)* 我 打掉了 杯子, 但是 杯子 不是 因为 我 打 它 才 翻的
 wǒ dǎ-diào-le bēizi, dànshi bēizi bùshì yīnwèi wǒ dǎ tā cái diào-de
 I hit-fall.PERF cup, but cup NEG because I hit it then fallen
 I hit-fell the cup, but the cup did not fall because I hit it.

(8)* 我 踢碎了 杯子, 但是 杯子 不是 因为 我 踢 它 才 碎的
 wǒ tī-suì-le bēizi, dànshi bēizi bùshì yīnwèi wǒ tī tā cái suì-de
 I kick-shatter.PERF cup, but cup NEG because I kick it then shattered
 I kick-shattered the cup, but the cup did not shatter because I kicked it.

Interpretation and Analysis

The data above show three main semantic patterns. First, simple action verbs such as 打 dǎ “hit” and 踢 tī “kick” do not by themselves entail a result state. In examples (1) and (2), the sentences remain acceptable even though the result state is denied. A speaker can say that they hit the cup without implying that the cup became broken, and they can say that they kicked the cup without implying that the cup became shattered. This means that 打 dǎ and 踢 tī should be analyzed as action predicates rather than result predicates. In lambda-calculus notation, they can be represented as transitive predicates such as:

打 dǎ = $\lambda y. \lambda x [\text{HIT}(x, y)]$ (a)

踢 tī = $\lambda y. \lambda x [\text{KICK}(x, y)]$ (b)

These representations only state that an action relation holds between an agent and a theme. They do not contain BROKEN(y), SHATTERED(y), or any other result-state predicate. This explains why denying the result state in examples (1) and (2) does not create a contradiction.

Second, the resultative compounds in examples (3) and (4) show that the result complement contributes an entailed result state. Once 破 pò “break” or 碎 suì “shatter” appears as the second element of the compound, denying the result state becomes contradictory or highly infelicitous. For example, 打破 dǎ-pò cannot naturally be followed by “but the cup did not break,” because the compound itself entails that the object becomes broken. Similarly, 踢碎 tī-

sui cannot naturally be followed by “but the cup did not shatter,” because 碎 sui contributes the shattered result state.

This supports a predicate-decomposition analysis. Instead of treating 打破 dǎ-pò and 踢碎 tī-sui as unanalyzed atomic predicates, the data suggest that they contain at least two semantic components: an action predicate contributed by the first verb and a result-state predicate contributed by the second verb. The result complements can be represented as:

$$\text{破 pò} = \lambda y[\text{BROKEN}(y)] \text{ (c)}$$

$$\text{碎 suì} = \lambda y[\text{SHATTERED}(y)] \text{ (d)}$$

The contrast between the simple action sentences and the resultative compound sentences shows that the result state does not come from 打 dǎ or 踢 tī alone. Rather, it is introduced by the second verbal element of the RVC.

Third, examples (5)–(8) show that Mandarin RVCs encode not only an action and a result, but also a causal relation between them. In examples (5) and (6), a simple action verb can be followed by a later result while denying that the action caused the result. For instance, it is possible to say that the speaker hit the cup, and the cup later broke, but that the breaking was not caused by the hitting. This is acceptable because the simple verb 打 dǎ only asserts the action of hitting; it does not assert that the hitting caused any particular result.

By contrast, examples (7) and (8) are contradictory or highly infelicitous. In these examples, the speaker uses a resultative compound such as 打掉 dǎ-diào or 踢碎 tī-sui and then denies that the result happened because of the action. This denial clashes with the meaning of the compound. The RVC already encodes that the action described by the first verb caused the result state described by the second verb. Therefore, the meaning of an RVC cannot be represented as a simple conjunction such as:

$$V(x,y) \wedge C(y) \text{ (e)}$$

This formula only says that an action happened and a result state occurred. It does not say that the action caused the result. The causation-denial examples show that a stronger representation is needed. A more accurate Dowty-style decomposition is:

$$V(x,y) \text{ CAUSE BECOME } C(y) \text{ (f)}$$

Under this analysis, 打破 dǎ-pò means that x's hitting of y caused y to become broken:

$$\text{打破 dǎ-pò} = \lambda y. \lambda x [\text{HIT}(x, y) \text{ CAUSE BECOME BROKEN}(y)] \text{ (g)}$$

Similarly, 踢碎 tī-suì means that x's kicking of y caused y to become shattered:

$$\text{踢碎 tī-suì} = \lambda y. \lambda x [\text{KICK}(x, y) \text{ CAUSE BECOME SHATTERED}(y)] \text{ (h)}$$

This analysis also explains how the compound can be built compositionally in lambda calculus. The first verb contributes a function from an object to a function from a subject to a proposition. The result complement contributes a function from an object to a result-state predicate. The RVC construction combines these two meanings and introduces the causal relation between them. A general rule can be stated as:

$$\text{RVC} = \lambda R. \lambda V. \lambda y. \lambda x [V(x, y) \text{ CAUSE BECOME } R(y)] \text{ (i)}$$

Combining (a) and (c) through the rule in (i) will yield (g).

To understand the lambda function for this sentence, we can first look at the lambda function of $V(x, y)$ which is $\langle e, \langle e, t \rangle \rangle$ (j) as it takes in a subject and a object. The lambda abstract version of $C(y)$ is $\langle e, t \rangle$ (k). Because the RVCs still function as a transitive verb, its own final function form is $\langle e, \langle e, t \rangle \rangle$. Because (i) takes in a $\langle e, \langle e, t \rangle \rangle$ and a $\langle e, t \rangle$ to form a $\langle e, \langle e, t \rangle \rangle$, its type is $\langle \langle e, \langle e, t \rangle \rangle, \langle \langle e, t \rangle, \langle e, \langle e, t \rangle \rangle \rangle \rangle$ as shown in figure 3.

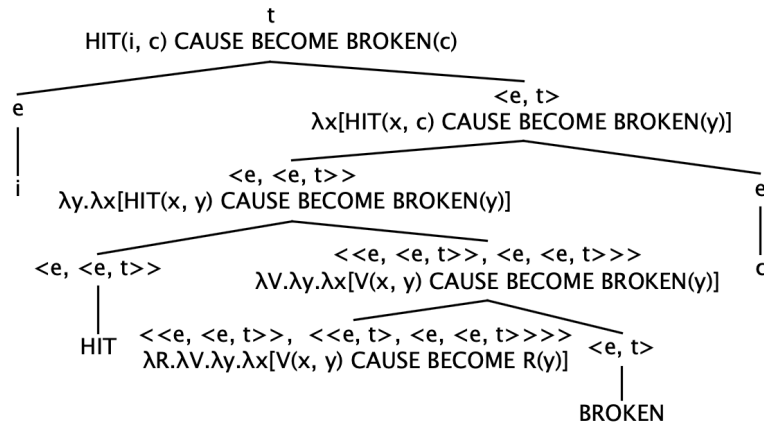


Fig 3

Conclusion

This paper has argued that Mandarin resultative verb compounds are best analyzed as semantically complex predicates rather than as unanalyzed atomic predicates. Focusing on

object-oriented RVCs formed with 打 dǎ “hit” and 踢 tī “kick,” the data show that the first verb contributes an action predicate, while the second verbal element contributes a result-state predicate. Simple action verbs such as 打 dǎ and 踢 tī do not entail that the object becomes broken, shattered, fallen, or otherwise changed. However, when these verbs combine with result complements such as 破 pò “break,” 碎 suì “shatter,” or 掉 diào “fall,” the resulting compound does entail that the object undergoes the relevant change of state.

The entailment and causation-denial diagnostics support a Dowty-style predicate-decomposition analysis. The meaning of a compound such as 打破 dǎ-pò should not be represented simply as an action plus an independent result, such as $HIT(x,y) \wedge BROKEN(y)$. Such a representation would fail to capture the causal relation encoded by the compound. Instead, the data support a representation in which the action caused the result: $HIT(x,y) \text{ CAUSE BECOME } BROKEN(y)$.

Lambda calculus provides a way to make this compositional structure explicit. The first verb can be represented as an action predicate, such as $\lambda y.\lambda x[HIT(x,y)]$, while the result complement can be represented as a result-state predicate, such as $\lambda y[BROKEN(y)]$. The RVC-forming operation combines these two pieces into a complex transitive predicate: $\lambda y.\lambda x[HIT(x,y) \text{ CAUSE BECOME } BROKEN(y)]$. This predicate then combines with the object and the subject to produce a complete sentence meaning. In this way, the compound can function syntactically like a single verb while still having a decomposable internal semantic structure.

Overall, Mandarin RVCs show how a compact verb compound can encode action, causation, and change of state at the same time. The analysis developed in this paper supports the view that resultative compounds are not merely lexicalized verb forms, but compositional structures whose meanings can be formally represented through predicate decomposition and lambda abstraction.

Bibliography

- Dowty, D. R. (1976). Montague grammar and the lexical decomposition of causative verbs. In *Montague grammar* (pp. 201-245). Academic Press
- Dowty, D. R. (1979). Word meaning and Montague grammar: The semantics of verbs and times in generative semantics and in Montague's PTQ (Vol. 7). Springer Science+Business Media.
- Dowty, D. R. (1979). Linguistic Evidence for the Two Strategies of Lexical Decomposition. In *Word Meaning and Montague Grammar: The Semantics of Verbs and Times in Generative Semantics and in Montague's PTQ* (pp. 235-293). Dordrecht: Springer Netherlands.
- Hashimoto, A. Y. (1966). Embedding structures in Mandarin. The Ohio State University.
- Li, C. (2007). Mandarin resultative verb compounds: Where syntax, semantics, and pragmatics meet. Yale University.
- Li, C. (2013). Mandarin resultative verb compounds: Simple syntax and complex thematic relations. *Language Sciences*, 37, 99–121. <https://doi.org/10.1016/j.langsci.2012.11.001>
- Li, Y. (1990). On VV compounds in Chinese. *Natural language & linguistic theory*, 8(2), 177-207.
- Lu, J. H. (1977). Resultative verb compounds vs. directional verb compounds in Mandarin. *Journal of Chinese linguistics*, 276-313.
- Rappaport Hovav, M., & Levin, B. (1998). Building verb meanings. In M. Butt & W. Geuder (Eds.), *The projection of arguments: Lexical and compositional factors* (pp. 97–134). CSLI Publications.
- Tham, S. W. (2009, March). Building resultatives in Mandarin (from the result). In 10th Semantics Fest (Stanford Semantics and Pragmatics Workshop), Stanford University.
- Thompson, S. A. (1973). Resultative verb compounds in Mandarin Chinese: A case for lexical rules. *Language*, 49(2), 361–379. <https://doi.org/10.2307/412459>
- Wang, W. S.-Y., & Sun, C. (Eds.). (2015). *The Oxford handbook of Chinese linguistics*. Oxford University Press.
- Wechsler, S., & Noh, B. (2001). On resultative predicates and clauses: Parallels between Korean and English. *Language Sciences*, 23(4–5), 391–423. [https://doi.org/10.1016/S0388-0001\(00\)00031-0](https://doi.org/10.1016/S0388-0001(00)00031-0)